

Reproduction of huggingface/lerobot: Diffusion Policy on PushT

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Independent reproduction · Artifact evaluation · Robotics

Abstract

Trained LeRobot's Diffusion Policy on the lerobot/pusht dataset at the pinned commit for 20,000 steps (batch 64, seed 1000, default diffusion settings), a tenth of the 200,000-step budget used for the published checkpoint, and evaluated it in the gym-pusht simulator on 50 episodes at 10,000 and 20,000 steps (success rate and maximum coverage reward). Evaluation ran in-process (use_async_envs=false) because the asynchronous evaluation workers could not load the simulator. One NVIDIA RTX 5060 Ti.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Diffusion Policy PushT (20k s...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: Diffusion Policy PushT (20k steps)

Run	Metric	Final value	Step
1	train_loss	0.007	20000
1	pc_success	36	20000
1	avg_max_reward	0.883922	20000

Run	Inline claim	Value	Phase
1	pc_success	36	
1	avg_max_reward	0.883922	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	4653.0 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	fa0a8f23ab3b9c32431901c7071efcfaac528842891dace8b1ab802263a28f53	7.6 MB
r1.kvbundle.zip	1fe7875e920c8fab4889ba1e2876a8a4882c3c46e47e384f87a078b78d9211e6	3.3 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2303.04137>

Reproduction of: <https://github.com/huggingface/lerobot/tree/ca69a2068462a37f7cdcb74180927a2f863d2bf7>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: diffusion-policy, imitation-learning, pusht, robotics.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Diffusion Policy PushT (20k steps)), kv:2610.00019v1/r1. Data hash 2bb6af12eec1fe3f8f3105ad97a4b514f4f339a90bd714cd0b017e830c077af8.
<https://kveritas.org/records/2610.00019v1/r1.pdf>